Multi-modal Large Language Models for Video Anomaly Detection

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Abstract

Video anomaly detection (VAD) seeks to temporally localize abnormal events in videos. Most existing methods rely on training deep models to learn normal behavior, which limits generalization and incurs high deployment costs due to the need for data recollection and retraining across domains.

We introduce OV-LAVAD, a completely training-free, language-based framework for VAD. It combines a vision-language model to generate frame-level captions and temporal summaries with a large language model guided by structured prompts to produce frame-level anomaly scores based on visual and contextual semantics.

Evaluated on the UCF-Crime dataset, OV-LAVAD demonstrates strong performance and surpasses several unsupervised and one-class baselines—without requiring any training or domain-specific adaptation.

1. Introduction

Video anomaly detection (VAD) is a fundamental problem in computer vision with broad applications in surveillance, public safety, traffic monitoring, and behavioral analysis. The goal of VAD is to temporally localize rare and unexpected events in long, untrimmed videos. A robust VAD system can enable real-time alerts, reduce manual inspection effort, and enhance situational awareness in dynamic environments—making it highly relevant for practical deployment.

Existing methods for VAD can be broadly categorized into three types. **Supervised approaches** treat anomaly detection as a binary classification or temporal segmentation problem, requiring extensive labeled training data. **One-class learning methods** learn the distribution of normal behavior and detect deviations at test time, under the assumption that anomalies do not resemble normal patterns. **Unsupervised techniques**, often based on reconstruction or prediction models (e.g., autoencoders or future frame prediction), assume that anomalies are harder to reconstruct or anticipate.

While these methods have shown progress, they all suffer from a core limitation: strong reliance on training data and domain-specific assumptions. Supervised models are expensive to annotate and struggle to generalize across scenes. One-class and unsupervised approaches reduce labeling cost, but are highly sensitive to what is implicitly considered ”normal,” and prone to misclassifying rare but benign events. Moreover, most existing methods require retraining when deployed in new environments, limiting their scalability and flexibility in real-world applications.

These challenges motivate a paradigm shift: can we build a training-free, generalizable framework for anomaly detection that requires no labeled data or task-specific retraining?

In this work, we propose **OV-LAVAD**, a novel training-free framework for video anomaly detection that departs fundamentally from conventional learning-based paradigms. Instead of relying on feature learning from labeled or normal-only data, OV-LAVAD leverages the reasoning capabilities of large language models (LLMs) guided by rich visual and temporal context generated by vision-language models (VLMs). This allows the system to detect anomalies without any task-specific supervision, model fine-tuning, or domain adaptation.

Specifically, for each frame in a video, we use a pretrained VLM (LLaVA-OneVision) to produce (1) a detailed caption describing the visual content of the frame, and (2) a temporal summary covering the 10-second window centered around the frame. These descriptions form a multimodal prompt that is passed to a powerful instruction-tuned LLM (LLaMA3-Instruct), which assigns a discrete anomaly score based on semantic reasoning. The entire pipeline is zero-shot and plug-and-play, requiring no training data, no parameter updates, and no prior knowledge of the deployment environment. This marks a shift toward language-native video understanding, where pretrained models can perform anomaly detection solely through prompt-based inference.

We evaluate our proposed OV-LAVAD framework on the

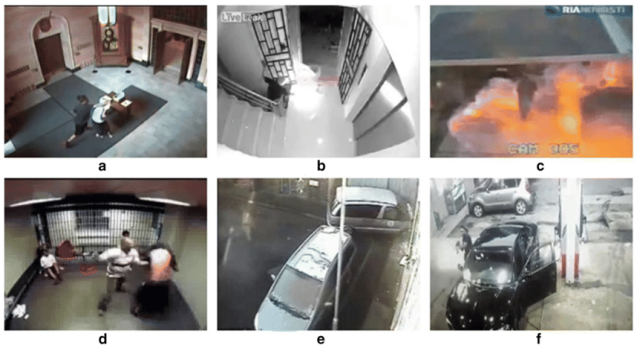


Figure 1. Example frames from the UCF-Crime dataset [11], illustrating various real-world surveillance scenarios such as fighting, road accidents, and burglary.

benchmark dataset UCF-Crime [11], and demonstrate that it consistently outperforms existing unsupervised and one-class VAD methods—despite requiring no training or data collection. These results provide strong empirical evidence that video anomaly detection can be effectively addressed in a fully training-free manner.

Our main contributions are summarized as follows:

- We propose **OV-LAVAD**, a fully training-free framework for video anomaly detection that eliminates the need for supervision, fine-tuning, or domain adaptation.
- We design a multimodal prompting strategy that combines frame-level captions and temporal summaries to guide large language models in generating anomaly scores through semantic reasoning.
- We conduct comprehensive experiments on the large-scale **UCF-Crime** dataset, demonstrating that OV-LAVAD achieves strong performance and outperforms several existing unsupervised and one-class baselines—despite using no training data.

2. Related Work

2.1. Traditional Methods for Video Anomaly Detection

Traditional methods for video anomaly detection (VAD) fall into three main categories: supervised, one-class, and unsupervised approaches.

Supervised methods [5, 11] treat anomaly detection as a binary classification or temporal localization task, requiring large-scale labeled datasets. While effective in controlled settings, their performance degrades significantly when deployed in unseen environments due to their reliance on class-specific training data and brittle decision boundaries.

One-class learning methods [10, 14] aim to model the distribution of normal behavior during training and detect anomalies as deviations from this learned manifold. These methods alleviate the need for explicit anomaly labels, but remain highly sensitive to visual context, camera viewpoints, and domain shifts.

Unsupervised approaches [2, 6, 7] typically rely on reconstruction or prediction mechanisms—such as autoencoders, LSTMs, or GANs—under the assumption that abnormal events are inherently harder to reconstruct or forecast. However, this assumption is not always reliable in practice: rare but benign events may produce high reconstruction errors, while subtle anomalies may appear normal to the model.

Despite their differences, these approaches all rely on training data and often require domain-specific fine-tuning. Their generalization capabilities remain limited, and performance tends to deteriorate when applied to scenes or behaviors not seen during training. These limitations motivate the exploration of training-free paradigms that can operate effectively across diverse, real-world environments.

2.2. Prompt-Based and Language-Driven Vision Methods

Recent advances in large language models (LLMs) and vision-language models (VLMs) have enabled a new paradigm of language-driven visual understanding. Models such as GPT-4 [8], LLaMA [12], and CLIP [9] have demonstrated strong capabilities in zero-shot classification, captioning, and retrieval by aligning vision and language in a shared embedding space. More recent VLMs such as BLIP-2 [3], Flamingo [1], and LLaVA [4] further extend these capabilities to perform open-ended reasoning on image or video content by generating natural language responses conditioned on visual inputs.

These models enable prompt-based interaction, where a carefully designed textual instruction can guide the model to perform a wide range of downstream tasks without fine-tuning. Prompt engineering has become a practical alternative to traditional training in tasks such as visual question answering [3], referring expression comprehension [13], and image captioning [3, 4].

While prompt-driven vision models have shown strong performance on recognition and understanding tasks, their use in **video anomaly detection** remains largely unexplored. A recent effort, LAVAD [15], explores the use of LLMs to score anomalies from captions and temporal context, but still incorporates multiple handcrafted filtering steps and memory-based refinement. In contrast, our proposed method **OV-LAVAD** uses a clean, modular pipeline where a VLM generates semantic-rich descriptions, and a frozen LLM directly assigns anomaly scores through instruction-based reasoning—without filtering, training, or post-hoc aggregation.

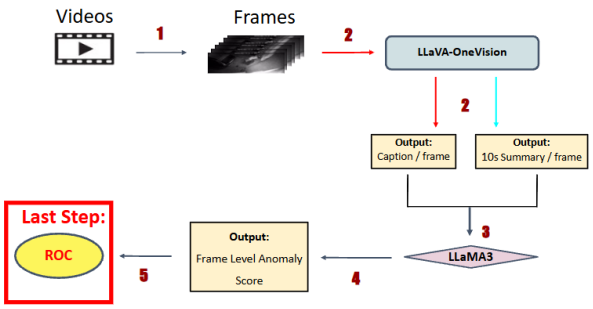


Figure 2. Overview of the OV-LAVAD pipeline. The model generates frame-level captions and temporal summaries, which are processed by an LLM to score anomalies.

2.3. Comparison to Our Work

In contrast to prior work, our approach introduces a fundamental shift in how video anomaly detection is framed. Traditional methods—whether supervised, one-class, or unsupervised—require training on normal or labeled data, operate under strong assumptions about visual distributions, and often suffer from poor generalization when deployed across domains. Even recent attempts to incorporate language, such as LAVAD [15], still rely on auxiliary components like similarity filtering, memory banks, or hand-designed rules to achieve usable anomaly scores.

Our proposed method, **OV-LAVAD**, departs from these strategies in three key ways. First, it is entirely training-free: we use frozen, off-the-shelf models without any adaptation or fine-tuning. Second, it operates in a fully modular pipeline, where visual understanding and anomaly reasoning are cleanly separated through a multimodal prompting interface. Third, it relies purely on semantic information—captions and temporal summaries—rather than low-level features or reconstruction loss. This enables plug-and-play deployment, domain-agnostic reasoning, and interpretable output, opening a new direction for anomaly detection through language-native video analysis.

3. Training-Free Video Anomaly Detection

We propose **OV-LAVAD**, a novel training-free pipeline for video anomaly detection that integrates pretrained vision-language and language models via structured prompting. Unlike existing approaches that require model training, domain adaptation, or anomaly-specific supervision, OV-LAVAD operates entirely at inference time using language as the medium for both visual understanding and anomaly reasoning. It is zero-shot, modular, and deployable without retraining.

3.1. Overview

Given those untrimmed videos from the UCF-Crime dataset, our goal is to assign an anomaly score to each frame, indicating the likelihood that abnormal behavior occurs at that timestamp. OV-LAVAD proceeds in two stages:

- **Context generation via VLM.** For each sampled frame, we employ a pretrained vision-language model to generate two forms of text: (i) a frame-level caption describing the scene, and (ii) a temporal summary representing the 10-second window centered around the frame.
- **Anomaly reasoning via LLM.** The caption and summary are embedded into a structured natural language prompt and passed to a pretrained instruction-tuned large language model. The model outputs a discrete anomaly score from a fixed scale (0.0 to 1.0), reflecting its semantic judgment of abnormality.

An illustration of our pipeline is provided in Figure 2. All components are off-the-shelf and frozen. No training or data collection is required, yet the system achieves strong anomaly detection performance across unconstrained real-world surveillance videos.

3.2. Visual and Temporal Context Generation

For each video, we uniformly sample frames at fixed intervals (e.g., one every 16 frames), and extract semantic context for each sampled frame using pretrained vision-language models. To capture both spatial and temporal semantics, we generate two forms of natural language descriptions per frame: a frame-level caption and a 10-second temporal summary.

Frame captions. We employ LLaVA-OneVision to process each frame independently and produce a zero-shot caption that describes the visual content of the scene. The resulting captions are stored in a JSON file per video, where each entry is indexed by the corresponding frame number.

Temporal summaries. To encode the broader scene dynamics, we generate a textual summary of a 10-second temporal window centered at each frame. Since LLaVA-OneVision is limited to a maximum of 8 image tokens, it cannot directly summarize a 10-second clip (typically 300 frames at 30 FPS). Instead, we first collect the captions of 300 frames surrounding the current frame, then use the LLM embedded in the Qwen2 architecture to produce a coherent semantic summary based on this caption set.

Formally, let f_i denote the i-th frame in a video, and let c_j denote the caption of frame f_j. For a target frame f_i, we define the set of surrounding captions as:

$$\mathcal{C}_i = \{c_j \mid j \in [i - \lfloor W/2 \rfloor,\ i + \lfloor W/2 \rfloor],\ 1 \leq j \leq N\} \quad (1)$$

where W is the window size (e.g., $W = 300$), N is the total number of frames in the video, and the index range is clipped to ensure valid frame access at the video boundaries. In practice, this corresponds to using fewer than 300 captions near the beginning and end of the video.

The temporal summary for the frame f_i is then generated by inputting $\mathcal{C}_i$ to Qwen2's LLM in natural language form (e.g., as a paragraph of concatenated captions), prompting it to return a concise summary of the scene during that period. These summaries are stored in a separate JSON file per video, aligned by frame index.

Together, the frame caption and temporal summary form a multimodal context that is used in the next stage to infer anomaly scores via large language model reasoning.

3.3. Language-Based Anomaly Reasoning

After obtaining frame-level captions and temporal summaries, we perform anomaly scoring using a large language model (LLM) guided by structured natural language prompts. For each frame, we construct a multimodal prompt that fuses the current visual description and its temporal context, then instruct the LLM to assign a discrete anomaly score ranging from 0.0 (normal) to 1.0 (highly abnormal), using fixed intervals of 0.1.

Initial prompt design. We began with manually written, human-interpretable prompts, instructing the LLM (LLaMA3-8B-Instruct) to analyze the caption and summary and provide an anomaly score. However, empirical results showed that such prompts often led to vague or inconsistent responses. This observation aligns with recent findings in prompt engineering, where prompt quality has a significant impact on LLM behavior.

LLM-guided prompt optimization. Inspired by Zhou et al. [16], who show that LLMs can serve as strong prompt engineers themselves, we reformulated our strategy. Instead of manually crafting prompts, we described our task objectives directly to the LLM and asked it to generate several candidate prompts optimized for semantic anomaly detection. Specifically, we queried LLaMA3-8B-Instruct with a meta-instruction outlining the desired behavior: "Given a frame-level visual caption and a temporal summary, produce a score from 0.0 to 1.0 representing how abnormal the current frame appears." The model responded with several candidate prompt templates, from which we selected and tested the most effective based on performance and consistency.

Scoring. The final prompt structure was then uniformly applied to all frames. Each prompt is fed to the LLM with the corresponding caption and summary inserted into the appropriate slots. The model outputs a discrete anomaly score from the set $\{0.0, 0.1, 0.2, ..., 1.0\}$, which is stored per frame and later used for evaluation against ground truth temporal annotations.

4. Experiments

4.1. Dataset and Evaluation Protocol

We evaluate OV-LAVAD on the test split of the UCF-Crime dataset [11], a widely used benchmark for video anomaly detection in real-world surveillance scenarios. The full dataset consists of 1,900 untrimmed videos across 13 categories of anomalous activities (e.g., robbery, fighting, accident). For evaluation, we follow the official split and use the 290 test videos, which cover a broad range of behaviors and environments. Each video is annotated with temporal intervals indicating when anomalies occur.

The task is framed as frame-level anomaly scoring, and we report the area under the receiver operating characteristic curve (ROC-AUC) as our primary metric, consistent with prior work [11, 15]. For each frame, OV-LAVAD generates an anomaly score between 0.0 and 1.0. Binary ground-truth labels are derived from the temporal annotations: frames within an anomaly interval are labeled positive, and the rest negative. ROC-AUC is computed over all frames across the 290 test videos.

Unlike prior methods that train models using normal-only or labeled data, OV-LAVAD operates entirely in a training-free and zero-shot setting. No train-test split is used for learning; the test set is processed in inference-only mode. Our results are thus directly comparable to existing unsupervised and one-class methods.

4.2. Implementation Details

OV-LAVAD is implemented using publicly available pretrained models without any fine-tuning. For vision-language processing, we use **LLaVA-OneVision** to generate frame-level captions, and the embedded LLM within **Qwen-2** to summarize temporal context. For final anomaly reasoning, we use the instruction-tuned **LLaMA3-8B-Instruct** model.

Frame sampling. Each video is sampled at a fixed interval of 16 frames. For each sampled frame, we generate:

- A frame-level caption using LLaVA-OneVision on the RGB image.
- A temporal summary of the 10-second window centered around the frame (approximately 300 frames). Due to LLaVA-OneVision's token limitations (max 8 frames per input), we first generate captions for the 300 surrounding frames and then pass these captions into **Qwen-2** to produce a high-level summary.

Prompt-based scoring. The caption and summary for each frame are inserted into a structured natural language prompt, which is then passed to LLaMA3-8B-Instruct. The model outputs a discrete anomaly score from the set $\{0.0,$

0.1, ..., 1.0}. Inference is performed in batches. All models remain frozen throughout; no gradient updates or task-specific fine-tuning are performed.

Storage. Outputs are saved in per-video JSON files. Each video has:

- a caption file indexed by frame number,
- a summary file indexed by frame (each entry covers the 10-second context),
- an anomaly score file with final frame-level predictions.

All experiments are run on NVIDIA A100 or H100 GPUs depending on model compatibility.

4.3. Quantitative Results

We compare OV-LAVAD against several recent training-free or zero-shot baselines on the UCF-Crime test set. Table 1 reports the frame-level ROC-AUC of our method alongside four representative baselines.

Table 1. Frame-level ROC-AUC (%) on UCF-Crime test set.

Method	**ROC-AUC (%)**
ZS CLIP	53.16
ZS ImageBind (Image)	53.65
ZS ImageBind (Video)	55.78
LAVAD [15]	78.32
OV-LAVAD (Ours)	**81.13**

OV-LAVAD achieves a ROC-AUC of **81.13%**, surpassing LAVAD [15], the strongest prior language-based baseline, by nearly 3 points. Compared to zero-shot vision-language embedding baselines such as CLIP and ImageBind, our method yields over 25 points of improvement. This highlights the strength of semantic-level reasoning over low-level feature matching.

These results confirm our central hypothesis: video anomaly detection can be effectively addressed using only textual descriptions and prompt-based inference, without requiring any training or adaptation. OV-LAVAD demonstrates that pretrained LLMs and VLMs, when properly composed, can serve as general-purpose, explainable anomaly detectors.

4.4. Evaluation Metric: ROC-AUC

We use the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) as the primary evaluation metric for frame-level anomaly detection. ROC-AUC quantifies a model's ability to distinguish between normal and abnormal frames across all possible threshold settings.

The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) at various decision thresholds:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{2}$$

Here, TP, FP, TN, and FN denote the number of true positives, false positives, true negatives, and false negatives, respectively. The ROC-AUC is computed as the area under this curve:

$$\text{AUC} = \int_0^1 \text{TPR}(t)\, d\text{FPR}(t) \tag{3}$$

A perfect classifier achieves an AUC of 1.0, while a random guess yields an AUC near 0.5. In our case, the model generates continuous frame-level anomaly scores, which are thresholded at varying levels to compute the ROC curve. As a threshold-independent measure, ROC-AUC is robust to class imbalance and widely adopted in anomaly detection benchmarks [15].

5. Conclusion

In this work, we propose **OV-LAVAD**, a novel training-free framework for video anomaly detection that operates entirely through language-based reasoning. OV-LAVAD leverages the synergy between pretrained vision-language models (VLMs) and large language models (LLMs) to estimate frame-level anomaly scores without any supervision or fine-tuning. Our pipeline consists of three main components: (1) a VLM is used to generate per-frame captions and temporal summaries based on a 10-second window centered around each frame; (2) these textual descriptions are formatted into structured prompts and passed to an instruction-tuned LLM to produce discrete anomaly scores for each frame; and (3) the resulting scores form the basis for temporal anomaly localization, evaluated at the frame level.

We benchmark OV-LAVAD on the challenging UCF-Crime dataset and show that our method achieves competitive or superior performance compared to existing unsupervised and one-class methods—despite requiring no training or data collection. Our results highlight the potential of multi-modal large language models as general-purpose, explainable tools for video understanding and open new directions for scalable anomaly detection systems.

However, while OV-LAVAD demonstrates the promise of training-free, language-based anomaly detection, a key limitation remains: the large language model has no explicit notion of what constitutes an "anomaly." Its scoring decisions are based entirely on surface semantics and commonsense priors, without a formal grounding in normal or abnormal behavior.

To address this, we plan to introduce **external memory mechanisms**, such as a memory bank of prototypical normal patterns. This will allow the LLM to compare each current scene against a reference distribution of normality, thus bridging the gap between language-based reasoning and task-specific definitions of anomaly.

In parallel, we also plan to explore **how providing richer and multi-level contextual information**—such as longer temporal summaries, motion trajectories, or surrounding frame cues—might further enhance the LLM's ability to make more accurate anomaly judgments. This line of work is orthogonal to the memory-based direction, aiming instead to boost the model's temporal reasoning and context sensitivity.

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References

[1] Jean-Baptiste Alayrac et al. Flamingo: a visual language model for few-shot learning. *NeurIPS*, 2022. 2

[2] Mahmudul Hasan et al. Learning temporal regularity in video sequences. In *CVPR*, 2016. 2

[3] Junnan Li et al. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *CVPR*, 2023. 2

[4] Haotian Liu et al. Visual instruction tuning. *arXiv preprint arXiv:2304.08485*, 2023. 2

[5] Wen Liu, Weixin Luo, Deyu Lian, and Shenghua Gao. Future frame prediction for anomaly detection–a new baseline. In *CVPR*, 2018. 2

[6] Weixin Luo et al. Revisit sparse coding: A robust unsupervised method for abnormal event detection in crowded scenes. In *CVPR*, 2017. 2

[7] Thao Minh Nguyen et al. Anomaly detection in video sequence with appearance-motion correspondence. In *ICCV*, 2019. 2

[8] OpenAI. Gpt-4 technical report, 2023. https://openai.com/research/gpt-4. 2

[9] Alec Radford et al. Learning transferable visual models from natural language supervision. *ICML*, 2021. 2

[10] Mohammad Sabokrou et al. Adversarially learned one-class classifier for novelty detection. In *CVPR*, 2018. 2

[11] Waqas Sultani, Chen Chen, and Mubarak Shah. Real-world anomaly detection in surveillance videos. In *CVPR*, 2018. 2, 4

[12] Hugo Touvron et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023. 2

[13] Jialu Yang et al. Unified vision-language pre-training for referring expression understanding. *CVPR*, 2023. 2

[14] Muzammal Zaheer et al. Old is gold: Redefining the adversarially learned one-class classifier training paradigm. In *CVPR*, 2020. 2

[15] Luca Zanella, Willi Menapace, Massimiliano Mancini, Yiming Wang, and Elisa Ricci. Harnessing large language models for training-free video anomaly detection. In *CVPR*, pages 18527–18536, 2024. 2, 3, 4, 5

[16] Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. Large language models are human-level prompt engineers. In *ICLR*, 2023. 4